

Fig. 4: Timing diagram of the proposed system

high-value resistor and 4700 μ F capacitor, and connect its output to the clock input of the CD4017. Assemble IC 555B in monostable mode with a 1M Ω resistor and 9 μ F capacitor. Connect the CD4017 outputs (Q0-Q6) to the

day-indicating LEDs through BC549 transistors.

Next, connect the optocoupler (PC817) to the monostable output through a current-limiting resistor and interface it with the BT66 audio module. Then connect a BC549 transistor and speaker. Ensure all grounds are common and connections are secure before applying a regulated 5V supply.

On powering the circuit, it begins operating automatically. An astable timer generates periodic pulses that advance the CD4017 decade counter IC outputs in sequence, lighting one LED at a time to indicate the current day.

Each pulse also triggers a monostable timer, producing an interval of around ten seconds during which the selected LED remains on and an audio alert is activated. After this period, the output turns off, and the system returns to standby, repeating the cycle daily and resetting after seven steps.

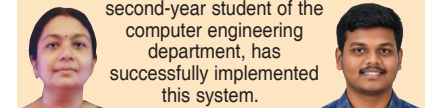
Based on this seven-day cycle, seven LEDs can be integrated into a weekly pillbox, as shown in Fig. 5. The pillbox is labelled from Sunday to Saturday, with each compartment corresponding to one LED. As the counter advances, the LEDs glow sequentially, indicating the active day, similar to how the pillbox guides daily medication.

This arrangement provides a simple visual indicator of weekly medication status. **EFY**



Fig. 5: Seven-day pillbox showing a seven-LED array for weekly status display

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